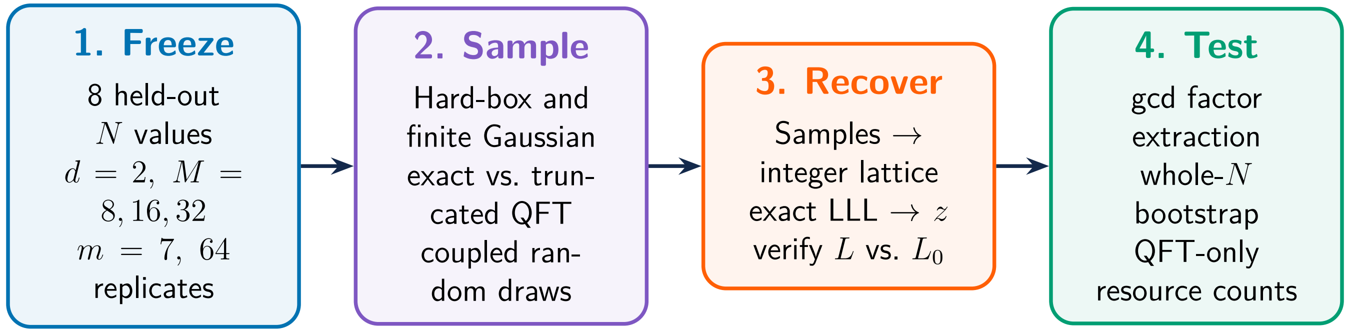
# Methods




**Research question.** Does removing the smallest controlled-phase rotations from the inverse quantum Fourier transform (QFT) preserve factor recovery at the complete finite Regev-style lattice endpoint?

1 **Freeze the test.** We fixed eight held-out semiprimes, $d = 2$, stored roots $(b_1, b_2) = (2, 3)$, Fourier moduli $M \in \{8, 16, 32\}$, $m = 7$ samples, 64 coupled replicates per cell, and master seed $2026071301$.

2 **Generate exact finite Fourier samples.** We evaluated both the notebook's uniform hard-box state and an exact finite discrete-Gaussian state with $R = 4$. Each condition used either the exact inverse QFT or a distance-truncated inverse QFT that removes the smallest-angle phase layers.

3 **Run the complete classical endpoint.** Measured samples were converted to an augmented integer lattice, reduced with exact LLL, and lifted to candidate relations $z$. We verified

$$\prod_i a_i^{z_i} \equiv 1 \pmod N, \qquad a_i = b_i^2 \bmod N,$$

then used the stored roots $b_i$ to distinguish useful $L \setminus L_0$ relations and extract factors with a gcd.

4 **Compare exact and truncated QFTs.** Exact and truncated conditions used coupled random draws. Factor-recovery differences were aggregated with $N$, not repeated trials, as the unit of generalization. A 5,000-draw whole-$N$ cluster bootstrap tested the frozen $-0.10$ non-inferiority margin. We separately counted QFT-only controlled-phase gates, transpiled CX gates, and depth.

Known factors were never used for sampling, tuning, or relation classification; they were used only for post-hoc validation.

### Frozen parameters

$N \in \{55, 65, 85, 95, 115, 119, 133, 161\}$
Stored roots: $(2, 3)$
$d = 2,\ M \in \{8, 16, 32\},\ m = 7$
Gaussian radius: $R = 4$
64 coupled replicates per cell
5,000 whole-$N$ bootstrap draws
Primary margin: $0.10$

### What this is not

This is a finite exact simulation study of QFT phase truncation. It is not calibrated hardware noise, cryptographic-scale factoring, or an end-to-end speedup benchmark.